Exploring Factors Influencing Fire Evacuation Wayfinding Confidence using Synthetic Respondent Surveys

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Abstract

Fire emergencies demand rapid wayfinding under stress and visibility constraints, where human behavior determines evacuation outcomes. This study examines demographic, psychological, and environmental determinants of wayfinding confidence (WFC)—a critical psychological precursor to effective egress—using 300 synthetic respondents conditioned on the General Social Survey. Findings reveal that higher wealth, education, employment, male gender, and marital status predict greater WFC, whereas older age diminishes it. Message consistency enhances WFC both directly and indirectly through alarm credibility mediation. Personal app guidance increases WFC, while reliance on social cues weakens individual confidence. Route familiarity moderates the effects of technological aids but not social factors, supporting a dual-process model that distinguishes cognitively mediated pathways (System 2: alarms, apps) from heuristic ones (System 1: social cues, familiarity). These results bridge engineering simulations with psychological frameworks, highlighting the need for targeted interventions—consistent messaging, app integration, and independent navigation training—to improve evacuation outcomes among vulnerable populations.

1 Introduction

Fire evacuation requires rapid interpretation of ambiguous cues and selection of egress routes under stress, time pressure, and visibility constraints. Human behavior during evacuation—particularly pre-movement delays and path choices—remains the primary determinant of casualties, often overriding structural safeguards (Proulx, 2001; Kuligowski, 2013). This study synthesizes these dynamics into a dual-process model, testing how alarm credibility, technological aids, social cues, and route familiarity shape wayfinding confidence—a critical psychological precursor to effective egress—using structural equation modeling (SEM) on population-representative synthetic survey data.

Recent work emphasizes its value for benchmarking synthetic agents' ability to reproduce human responses (Park et al., 2024). This study utilized synthetic survey respondents generated from the General Social Survey (GSS)-based agent bank in the Stanford HCI "genagents" framework to collect data for hypothesis testing. Each synthetic respondent, representing a fictional individual with sociodemographic attributes (e.g., age, gender, race, education, household characteristics) sampled to match GSS joint distributions, was instantiated as a Large langue model (LLM)-driven agent; these attributes were provided as conditioning context in the agent's system and memory prompts to elicit realistic responses to a custom evacuation wayfinding survey instrument. The survey was administered to 300 agents with pre-registered templates and fixed model settings.

Existing literature confirms interactive determinants of evacuation decision-making but lacks SEM testing of causal mediation (e.g., message consistency via alarm credibility) and moderation effects (e.g., familiarity × app guidance). Individual risk perception, social influence, environmental factors, and low alarm credibility drive delays, with wayfinding confidence as a key underexplored mediator. Empirical integration of mobile technologies with cognitive moderators remains sparse. No prior studies have unified these factors into a testable framework using diverse, GSS-conditioned synthetic respondents to benchmark population patterns without real-participant ethical constraints. Addressing these gaps, the current study tests eight hypotheses derived from questionnaire syntheses and experiments.

Findings advance egress modeling by validating cognitive (System 2: alarms/apps) versus heuristic (System 1: social/familiarity) pathways, bridging engineering simulations with psychological frameworks. Demographics (wealth, education, employment), message consistency (direct/via alarm credibility), and personal app guidance significantly enhance wayfinding confidence during evacuations, while social cue reliance reduces it. Route familiarity strengthens technological aid effects but not social factors, advocating targeted interventions for vulnerable groups through consistent messaging, app integration, and independent skills training.

2 Literature review

2.1 Fire evacuation

2.1.1 Decision-making factors

This section synthesizes previous studies on factors influencing fire evacuation decision-making—initiation, delay, compliance, and route/exit choices. The decision factors analyzed in prior studies are categorized by theme. A summary of previous studies is presented in Table 1. Evacuation decision-making in fires emerges from interacting individual, social, environmental/building, information/communication, and organizational/policy determinants. Across contexts, raising perceived risk appropriately, clarifying cues, leveraging positive social influence, improving message design, and conducting realistic drills can reduce pre-movement delays and increase compliance (Fork, 2019; Vaiciulyte, 2020; Alhuthali, 2021; Wong et al., 2023; Suparto and Erwandi, 2024; Kinkel et al., 2025). Findings from prior studies on social influence and pre-movement variability is strong; evidence for the causal impact of organizational policies and demographic modifiers is mixed and often correlational. Many studies are lab-based or retrospective, limiting causal inference and generalizability across diverse communities (Folk, 2019; Wong et al., 2023; Kinkel et al., 2025).

Table 1: Summary of previous studies on factors influencing fire evacuation decision-making

Theme	Specific determinants	Effective interventions	Sources
Individual	Risk perception, Prior experience, Demographics, Health/panic/ attachment-seeking	Target risk messaging, Preparedness campaigns, Support for vulnerable	Folk(2019); Suparto and Wong et al.(2023); Erwandi(2024); Kinkel et al.(2025)
Social/ household	Social influence/ bystander behavior, Household consensus & family responsibilities, Peer actions and local social norms	Use of trusted local messengers, Community leader prompts, Group-focused drills	Vaiciulyte(2020); Alhuthali(2021)
Environmental / building	Layout & familiarity with routes, Egress availability, Visibility(smoke)/ physical geometry causing congestion	Improve signage & intelligible voice alarms, Staff-assisted egress, Design to reduce choke points, Incorporate building-specific messaging	Folk(2019); Alhuthali(2021)

Information/ communication	Alarm clarity & credibility, Warning timing & explicitness, Message source and content (urgency, actionability), Multi-channel alerts (social media, cell)	Clear & specific action messages, Multi-channel dissemination, Behavioral message inputs, Credible senders & hazard naming	Vaiciulyte (2020); Wong et al(2023); Suparto and Erwandi(2024)
Organizational / policy	Evacuation order type (mandatory vs voluntary), Training/drills/ job aids, Leadership, Safety climate, Evacuation planning	Regular & realistic drills, Job aids, Clear authority messaging, Pre-incident planning	Vaiciulyte (2020); Suparto and Erwandi(2024); Kinkel et al.(2025)
Pre-movement timing	Cue perception & ambiguity, Engaging tasks (collecting items)/ maintenance tasks, Initial position & congestion risk	Reduce ambiguity in cues, Design alarms & messages to minimize engaging tasks	Alhuthali(2021); Kinkel et al.(2025)

2.1.2 Wayfinding confidence and fire evacuation

Elevated anxiety and time pressure degrade decision quality and route choice, increasing errors and random choices (Yang, 2015; Iftikhar et al., 2021; Altıntaş, 2023). Previous studies show the calming and performance-enabling role of confidence-supporting guidance. Broader navigation literature compiled in a self-efficacy/spatial-anxiety compendium shows spatial anxiety negatively correlates with navigation accuracy and route learning, whereas higher wayfinding self-efficacy and sense of direction predict better shortcut-finding and route-learning—constructs closely tied to decision quality during wayfinding (Altıntaş, 2023). Disorientation increases stress and frustration, aligning with the interpretation that environmental aids mitigate stress and uncertainty (Iftikhar et al., 2021). Specificity and reliability of guidance (directional arrows vs detailed route instructions), presence of explicit negation (blocked exit symbols), and pre-exposure/training; more specific and credible information promotes decisive route following (Yang, 2015; Galea et al., 2014; Augustine, 2023). Crowd cues/herding under ambiguity, neighbors' behavior, perceived risk, and smoke level. Ambiguity increases reliance on others, whereas personal information or training reduces blind following. (Augustine, 2023). Familiarity from repeated exposure and VR/serious-game training associate with reduced stress and improved wayfinding performance, building self-efficacy transferable to emergencies (Lin et al., 2019; Iftikhar et al., 2021).

Collectively, confidence-supporting wayfinding affordances—clear, dynamic, and salient cues—reduce physiological stress and decisional hesitation in escape tasks, thereby enabling sounder decisions (Kostakos et al., 2021; Galea et al., 2014). To cultivate wayfinding confidence and support calmer decision-making during fire evacuation, it is essential to provide highly detectable, unambiguous, and dynamic guidance at decision points, enhance environmental legibility and landmark salience, deliver credible and specific instructions to minimize ambiguity, and account for individual and social differences by offering redundant, multimodal cues and regular drills to build self-efficacy (Galea et al., 2014; Yang, 2015; Iftikhar et al., 2021; Kostakos et al., 2021; Augustine, 2023).

2.2 Hypotheses development

Previous studies on fire evacuation decision-making have primarily focused on factors such as pre-evacuation delay, alarm interpretation/trust, information sources, social influence, familiarity and signage, and path/exit selection factors (Gerges et al., 2018; Magnusson and Pagnon, 2019; Haydock, 2000; Kinateder and Warren, 2016; Glauberman, 2018). Because the key decision-making stages that determine actual casualties and evacuation efficiency are "how residents respond to the alarm" and "which route they choose," a survey-based study examining residents' responses to fire alarms and their route selection behavior was reviewed. This study extends prior questionnaire-informed findings on residents' alarm response and path choice to propose testable hypotheses that address unresolved gaps identified in earlier studies.

2.2.1 Alarm credibility

Alarm credibility strongly affects evacuation response. Conflicting/changing alarm messages undermine trust and compliance (Kuligowski and Hoskins, 2010). A reconciliatory follow-up that explicitly references the prior instruction restores trust and increases compliance relative to a follow-up without reference (Kuligowski and Hoskins, 2010). Moreover, ambiguous tone-only alarms are often not recognized as a fire alarm; one compilation reports that only about one in five recognized a general ring as indicating fire, whereas spoken/voice messages were more readily identified and aided the decision to evacuate (May, 2016). Drawing on the reviewed literature, the following hypotheses are proposed regarding factors influencing alarm credibility and wayfinding confidence:

H1a: Message consistency of alarms influences evacuees' wayfinding confidence.

H1b: Alarm credibility mediates the effect of message consistency on evacuees' wayfinding confidence.

2.2.2 Path guidance and social cues influence

Path guidance aids, encompassing integrated mobile apps, must function effectively under smoke-obscured visibility and dynamic hazard conditions to enable adaptive evacuation. The hypothesis is presented as follows:

H2a: Personal app guidance influences evacuees' wayfinding confidence.

Social cues influence shape evacuees' wayfinding confidence during residential fire evacuation. Initial exit choices are more socially driven (following others) (Lovreglio et al., 2022). Route choice is shaped by perceived risk, leadership, disability/fitness, and social attachment (Alhuthali, 2021). When bystanders remain passive, individuals delay or refrain from acting; larger groups increase decision difficulty (Alhuthali, 2021). Drawing on the reviewed literature, the following hypothesis is proposed:

H2b: Social cues influence evacuees' wayfinding confidence.

2.2.3 Route familiarity

Previous studies consistently indicate that social cues, emotions, and building features guide wayfinding decisions (May, 2016; Alhuthali, 2021; Gerges, 2020; Cisek and Orłowska, 2024). Route choice is shaped by perceived risk, leadership, disability/fitness, and social attachment, often leading to avoidance of lifts and reliance on known stairs (Alhuthali, 2021). Occupants tend to choose familiar exits (often the main entrance/stair) and to follow other people, even when nearer exits exist (Gerges, 2020). Unfamiliar occupants reported more negative feelings and longer reaction times, indicating emotional state and familiarity affect route decisions (May, 2016; Alhuthali, 2021; Gerges, 2020; Cisek and Orłowska, 2024). Mid-evacuation hazard updates (e.g., smoke increase near chosen exit) shift decision weight toward hazard avoidance, increasing rerouting probability and speed, but route familiarity moderates this shift (Lovreglio et al., 2022). The hypotheses are presented as follows:

H3a: Route familiarity moderates the relationship between message consistency and alarm credibility, wherein the relationship is weaker when route familiarity is high.

H3b: Route familiarity moderates the relationship between message consistency and evacuees' wayfinding confidence, wherein the relationship is weaker when route familiarity is high.

H3c: Route familiarity moderates the relationship between personal app guidance and evacuees' wayfinding confidence, wherein the relationship is stronger when route familiarity is high.

H3d: Route familiarity moderates the relationship between social cues and evacuees' wayfinding confidence, wherein the relationship is weaker when route familiarity is high.

Figure 1 illustrates the hypothesized relationships among factors influencing evacuees' wayfinding confidence.

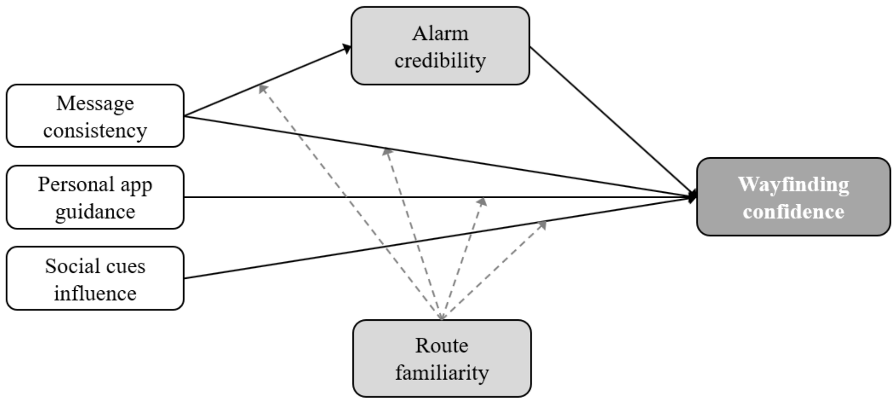

Figure 1: Conceptual model of hypotheses on evacuees' wayfinding confidence.

3 Method

3.1 Synthetic respondent surveys

This study used LLM–driven generative agents as synthetic survey respondents, using the publicly available demographic agent bank released with the Stanford HCI "genagents" framework (Park et al., 2024), specifically the population files which are derived from the General Social Survey (GSS) demographic distributions. The GSS is a long-running, nationally representative survey of U.S. adults with a stable core and rotating modules (Park et al., 2024). The GSS collects comprehensive information on respondents' demographic backgrounds, socioeconomic characteristics, social and political attitudes, and behavioral patterns, making it an ideal source for generating diverse, realistic synthetic agents.

Each record in this agent bank encodes basic sociodemographic attributes (for example, age, gender, race, education, and other household-level characteristics) sampled to approximate the joint distributions of the GSS reference population, while all personal identifiers (such as names and addresses) are fully fictionalized. For the purposes of this study, each GSS-based agent was treated as a synthetic respondent, and an LLM-driven agent was instantiated for every record, using its stored demographic attributes as conditioning context in the agent's system- and memory-level prompts to guide subsequent survey responses. Following prior work on generative agents, we conditioned each agent with its sociodemographic attributes profile (age, gender, race, education, and household characteristics) in the prompts to elicit context-consistent responses. This approach draws on evidence that agents informed by rich, person-centered context can replicate survey responses and subgroup patterns anchored to GSS items (Park et al., 2024; Anthis et al., 2025; Bisbee et al., 2024).

One example of the sociodemographic attributes set used to build the agent is listed in Table 2. Modules from the Genagents repository were utilized to instantiate generative agents conditioned on the specified sociodemographic attributes. Agents were queried through the framework's numerical_resp() interface for survey administration. For each of the 300 synthetic agents, the survey was administered via structured prompting, capturing responses to all measured constructs (message consistency, alarm credibility, wayfinding confidence, personal app guidance usage, social cues, and route familiarity) on validated Likert scales adapted from prior evacuation research.

Table 2: Example of sociodemographic attributes in a population dataset

Attribute	Value	Attribute	Value
First name	Thomas	Family structure at 16	Lived with parents

Last name	Collins	Family income at 16	Below average
Age	47	Father's highest degree	High school
Sex	Male	Mother's highest degree	Less than high school
Ethnicity	Austria	Mother's work history	Yes
Race	White	Marital status	Married
Detailed race	White	Work status	Retired
Hispanic origin	Not Hispanic	Military service duration	Yes, more than 4 years
Street address	742 Evergreen Terrace	Religion	Protestant
City	Bowling Green	Religion at 16	Protestant
State	KY	Born in U.S.	Yes
Political views	Slightly conservative	U.S. citizenship status	A U.S. citizen
Party identification	Not very strong republican	Highest degree received	Associate/junior college
Residence at 16	East South Central	Speak other language	No
Same residence since 16	Different state	Total wealth	Less than $5,000

3.2 Measures

Six latent constructs were measured using a total of 26 items on a 5-point Likert-type scale ranging from 1 (strongly disagree) to 5 (strongly agree). Message Consistency (MC) was assessed with 4 items evaluating the perceived consistency of emergency alarm messages. Alarm Credibility (AC) was measured using 4 items gauging the perceived trustworthiness and reliability of the alarm system. Wayfinding Confidence (WFC) was evaluated with 5 items measuring participants' confidence in their ability to find their way during emergency evacuations. Personal App Guidance (PA) was assessed using 5 items evaluating preferences for navigation app features aiding wayfinding. Social Cues (SO) was measured with 4 items assessing the tendency to follow others' behavior in emergency situations. Finally, Route Familiarity (RF) was evaluated using 4 items measuring participants' familiarity with evacuation routes.

3.3 Analytical strategy

The analysis proceeded in four stages, using KOMOS AI tool (python semopy library). First, demographic characteristics of the sample were examined using descriptive statistics including means, standard deviations, frequencies, and percentages. Second, multiple regression analysis was conducted to examine the simultaneous effects of demographic variables on WFC, controlling for all other predictors. Third, structural equation modeling (SEM) with a path analysis approach was employed to test the hypothesized relationships. Model fit was assessed using chi-square statistics, the comparative fit index (CFI), Tucker-Lewis index (TLI), and root mean square error of approximation (RMSEA). Acceptable fit was defined as CFI ≥ .90, TLI ≥ .90, and RMSEA ≤ .08 (Hu & Bentler, 1999; Kline, 2016).

4 Results

4.1 Preliminary Analyses

The sample included 300 participants with a mean age of 47.52 years (SD = 17.76) (Table3). Age distribution showed that 20.0% of participants were classified as young (18-30 years), 31.0% as middle-young (31-45 years), 22.3% as middle-old (46-60 years), and 26.7% as old (61+ years). This age distribution indicates a relatively balanced representation across different life stages, with slightly more representation from the middle-young and older adult categories. Regarding gender, the sample was predominantly female (56.3%, n = 169) compared to male (43.7%, n = 131). The racial composition of the sample was primarily White (66.3%, n = 199), followed by Black (17.0%, n = 51), and other races (16.7%, n = 50). The predominance of White participants reflects common patterns in U.S.-based survey research. In terms of wealth, participants were distributed across low (less than $20,000; 32.7%), middle ($20,000-$75,000; 37.3%), and high (greater than $75,000; 30.0%) categories.

To examine the influence of demographic characteristics on wayfinding confidence (WFC), multiple regression analyses revealed several significant predictors (Table 4). Full-time employment emerged as the strongest predictor (β = 0.767), with full-time workers reporting 0.453 points higher WFC than non-workers. Higher education level (β = 0.313) independently predicted greater WFC. Married status (β = 0.554) was associated with 0.327 points higher WFC, and greater wealth level (β = 0.321) also predicted increased WFC. In contrast, older age had a negative effect (β = -0.144), males reported 0.104 points higher WFC (β = 0.177), and part-time employment was also significant (β = 0.308).

Table 3: General demographic characteristics (N=300)

Variables	Categories	N	%
Age (years)	18-30	60	20.0
	31-45	93	31.0
	46-60	67	22.3
	61+	80	26.7
Gender	Female	169	56.3
	Male	131	43.7
Race	White	199	66.3
	Black	51	17.0
	Other	50	16.7
Education level	Less than High School	21	7.0
	High School	148	49.3
	Associate/Junior College	28	9.3
	Bachelor's	58	19.3
	Graduate	45	15.0
Total wealth	< $20K	98	32.7
	$20K - $75K	112	37.3
	> $75K	90	30.0

Table 4: Multiple regression results for demographic predictors of wayfinding confidence

Predictor	b	SE	β	t	p
Age	-0.005	0.002	-0.144	-2.74	.007**
Gender (Male)	0.104	0.042	0.177	2.47	.014*
Race (Black)	-0.036	0.057	-0.061	-0.63	.529
Race (Other)	-0.094	0.057	-0.158	-1.63	.104
Education Level	0.148	0.018	0.313	8.28	< .001***
Married	0.327	0.053	0.554	6.19	< .001***
Divorced	0.132	0.070	0.224	1.89	.060
Widowed	0.079	0.101	0.134	0.79	.433
Separated	0.005	0.107	0.009	0.05	.961
Full-time	0.453	0.057	0.767	7.95	< .001***
Part-time	0.182	0.072	0.308	2.52	.012*
Retired	0.046	0.079	0.078	0.58	.560
Wealth Level	0.221	0.026	0.321	8.54	< .001***

4.2 Hypothesis Testing

Table 6 summarizes hypothesis testing results. Six of eight hypotheses were supported. Message consistency influenced WFC both directly (b = 0.227) and through alarm credibility (indirect, b = 0.556). Personal app guidance significantly enhanced WFC (b = 0.409), highlighting navigation apps' positive role. Social cues influence negatively impacted WFC (b = -0.281), suggesting that dependence on others undermines personal confidence. Route familiarity weakened the MC-AC relationship (b = -0.294). Familiar individuals rely less on consistent messaging to trust alarms. Consistent messaging benefits confidence regardless of route familiarity. Route familiarity strengthened the PA-WFC relationship (b = 0.135). Navigation apps are most effective when users have existing environmental knowledge. The negative effect of social cues on confidence is consistent regardless of familiarity. CFA results demonstrated acceptable model fit (CFI = .92; TLI = .91; RMSEA = .059), with all factor loadings significant ($p < .05$). All models exhibited acceptable fit indices.

Table 6: Summary of hypothesis testing results

Hypothesis	Path	b	SE	p	Result
H1a	MC → WFC	0.227	0.107	.033*	Supported
H1b	MC → AC → WFC	0.556	0.076	< .001**	Supported
H2a	PA → WFC	0.409	0.043	< .001**	Supported
H2b	SO → WFC	-0.281	0.053	< .001**	Supported
H3a	MC × RF → AC	-0.294	0.084	< .001**	Supported
H3b	MC × RF → WFC	0.140	0.105	.183	Not Supported
H3c	PA × RF → WFC	0.135	0.049	.006**	Supported
H3d	SO × RF → WFC	-0.019	0.061	.755	Not Supported

5 Discussion

Demographic characteristics substantially influenced wayfinding confidence, with higher wealth, education, and full-time employment linked to greater confidence due to enhanced resources and navigation experience. Males and married individuals showed higher confidence, while older adults exhibited lower confidence consistent with age-related spatial cognition declines. Message consistency positively influenced wayfinding confidence both directly and indirectly through alarm credibility, highlighting the critical role of consistent communication in fostering trust and confidence during emergencies. Personal app guidance further enhanced wayfinding confidence, emphasizing the value of integrating navigation technologies into evacuation systems. In contrast, reliance on social cues negatively affected confidence, indicating that dependence on others may diminish individuals' perceived control and self-efficacy in emergency navigation.

Route familiarity reduced the impact of message consistency on alarm credibility, suggesting familiar users depend less on consistent messaging for trust formation. It did not moderate the direct link between message consistency and wayfinding confidence but strengthened the positive effect of personal app guidance, indicating apps are most effective for users with prior environmental knowledge. Route familiarity moderated the effects of technological wayfinding aids such as personal app guidance but not social factors, enhancing the benefits of structured guidance systems for familiar users. Specifically, strengthening the positive influence of personal app guidance on wayfinding confidence. These findings support a dual-process model of evacuation behavior, in which route familiarity facilitates cognitively mediated (System 2) processing of technological cues, whereas social cue following operates through heuristic (System 1) processes that are less dependent on spatial knowledge. This integration of dual-process theory into fire evacuation research bridges engineering-based and psychological perspectives and underscores the value of evacuation drills that increase route familiarity to improve the effectiveness of technological navigation aids.

5.1 Theoretical and Practical Implications

This study offers several theoretical contributions to the literature on emergency evacuation behavior. It demonstrates that demographic and socioeconomic factors—such as wealth, education, and employment—significantly shape wayfinding confidence, reflecting differences in life experience and access to resources. The findings also reveal that message consistency affects confidence both directly and indirectly through alarm credibility, reinforcing theories that emphasize consistent communication in building trust during emergencies. Moreover, the negative impact of social cue reliance suggests that depending too heavily on others can weaken individual self-efficacy, extending social influence theory to emergency contexts. The moderating effect of route familiarity indicates that communication and guidance strategies should consider individuals' prior environmental knowledge. This pattern reveals a dual-process mechanism in evacuation wayfinding. Technological wayfinding is cognitively mediated and enhanced by spatial knowledge, whereas social wayfinding is heuristic-based and less dependent on prior route knowledge.

The study presents practical implications for emergency management and building design. Emergency preparedness should prioritize groups with lower wayfinding confidence by providing clearer signage, better guidance systems, and targeted training. Consistent messaging across communication platforms is essential to build alarm credibility and trust, which enhance navigation confidence. Integrating personal navigation applications into evacuation systems can improve real-time guidance through familiar smartphone interfaces. Training should emphasize independent navigation rather than reliance on social cues, as social influence negatively affects evacuation effectiveness. Finally, practitioners and planners should consider users' route familiarity, offering alternative guidance (signage or staff assistance) for those less familiar with the environment.

5.2 Limitations and future directions

Several limitations and future research directions are noted. Given known issues with LLM-based synthetic respondents, the study focused on relative path patterns rather than precise estimates. Three constructs (MC, AC, SO) showed moderate reliability ($\alpha = .60–.64$), which is acceptable for exploratory research (Nunnally, 1978; Hair et al., 2010; Robinson et al.,1991) but may have weakened observed relationships. Data were self-reported rather than behavioral, restricting causal inference. Route familiarity was also self-reported, and effects may differ by context or population. Future studies should adopt experimental or longitudinal designs, include behavioral observations, and test the generalizability of findings across emergency types, building settings, and cultural contexts. Interventions that enhance wayfinding confidence among vulnerable groups and explore interactions between personal and environmental factors are recommended.

6 Conclusion

This study examined the demographic and psychological factors influencing wayfinding confidence (WFC) during emergency evacuations. Using a sample of 300 synthetic respondents, multiple regression analysis and SEM are employed to test a comprehensive model of wayfinding confidence determinants. In conclusion, this study provides evidence that wayfinding confidence during emergency evacuations is shaped by a complex interplay of demographic, psychological, and environmental factors. Demographic factors, particularly socioeconomic status and employment, explain a substantial portion of variance in wayfinding confidence. Message consistency enhances confidence both directly and through building alarm credibility, while personal app guidance offers a promising avenue for enhancing evacuation preparedness. The negative effect of social cues influence suggests that promoting independent navigation skills may be more beneficial than encouraging reliance on others. Route familiarity moderates several key relationships, highlighting the need for tailored intervention strategies. These findings underscore the importance of adopting a comprehensive approach to emergency preparedness that considers individual differences in wayfinding confidence. By targeting vulnerable populations, ensuring message consistency, leveraging personal navigation technology, and fostering independent navigation skills, emergency managers can enhance evacuation outcomes and save lives during critical events.

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